AURA - Stage 5: CERA Adaptive Inspection Planner

Technical implementation and controlled pilot report | October 24, 2026 | v1.0

Executive result

Stage 5 integrates an adaptive, counterexample-guided read-only inspection policy (CERA) with AURA Stages 1-4. It compares five methods on **240 authored finite-world episodes** (1,200 policy evaluations) and on **six real local SQLite/filesystem captures** (30 additional policy evaluations). The complete regression suite passes **153 automated tests**. All 30 local evaluations match independently authored scenario expectations.

The central result is **modest**: under the uniform finite-world benchmark, CERA averages **3.630 normalized marginal inspection cost units**, compared with **3.710 for fixed order**, **3.792 for greedy fact coverage**, and **3.597 for the exact small-instance optimizer**. An exhaustive checker uses **30.273** units. The improvement over fixed order is only **2.15%** and must not be characterized as significant without stronger evidence. These cost units are modeling assumptions, not milliseconds.

1. System architecture and correctness boundary

Previous components are reused rather than reimplemented: Stage 1 formal contracts and JSON schemas, Stage 2 independently captured state and read-only probes, Stage 3 contract verification and mutation frame checks, and Stage 4 bounded possible-world reasoning. Stage 5 adds aura_planning/planner.py, examples.py, and live_runner.py.

Input: A reviewed task contract, predeclared bounded model, observation permissions and estimated probe costs, and only the evidence already independently collected.

Output: Selected read-only probes, a Stage 4 verified PASS/FAIL/UNCERTAIN certificate, the trace of inspections, model-relative reasoning records, and actual/estimated cost accounting.

Key separation: In the synthetic simulator, only the observation oracle receives the hidden actual-world index; the selection API plan_next(...) does not. For real local captures, the initial-state S0 mark (70) is admitted into the finite model before any final-state inspection; the final state S_T and expected scenario label are not passed to the planner.

PASS safety: Even when all currently possible worlds look successful, AURA requires independent complete-frame coverage of every declared mutable surface. A known violation can be FAIL without inspecting unrelated surfaces. Inconsistent evidence, model undercoverage and unreachable outcomes return UNCERTAIN, never vacuous PASS.

2. Implemented policy baselines

• **Fixed early-stop:** Registered inspection order; stop on a decided verdict.

• **Cost-greedy early-stop:** Greedy marginal unresolved fact and coverage benefit per estimated unit cost.

• **CERA:** Adaptive scoring of pairs of still-compatible successful/failing worlds separated by each probe, normalized by cost; mandatory frame coverage remains explicit.

• **Exact bounded:** Dynamic programming maximizes chance of a decisive verdict, then minimizes expected marginal cost. This is only for small finite models (maximum 9 probes and 500 worlds).

• **Exhaustive:** Executes every available probe, even after an earlier conclusive result; a costly reference, not a strong competitor.

All methods receive the same catalogue, model assumptions, costs, evidence privileges, and contract. No policy is allowed to issue PASS solely from an agent status message.

3. Finite-world pilot results

Authored test set: Four domains (records, orders, access, files), four variants/domain and 12-16 enumerated worlds/variant = **240** world episodes, each run with all five policies. Four required boolean obligations include one mutation-frame obligation; six permitted probes include both redundant combined probes and a full-frame inspection. These are structurally similar artificial tasks, not 240 independent real-world agent workflows.

Policy	Mean marginal cost	Mean probes	Decisive coverage	False PASS	False FAIL
fixed	3.710	1.93	100%	0	0

Policy	Mean marginal cost	Mean probes	Decisive coverage	False PASS	False FAIL
greedy	3.792	1.80	100%	0	0
cera	3.630	1.93	100%	0	0
optimal	3.597	1.80	100%	0	0
exhaustive	30.273	6.00	100%	0	0

Interpretation: The small model gives all algorithms exact deterministic probe outcomes and a model that contains the actual test world. This makes zero false decisions an engineering soundness check, not proof of general AI-agent reliability. The high fraction of failed synthetic worlds makes early stopping unusually attractive. Estimated marginal cost omits the shared initial snapshot cost; the local integration reports that separately.

4. Stress testing and negative results

We reweighted the same worlds with an explicit public prior giving successful worlds 50% or 90% total probability. There was no change in the actual model or probe catalogue.

Policy	Expected cost, clean prior 0.5	Expected cost, clean prior 0.9
fixed	7.794	11.559
greedy	7.715	11.543
cera	7.715	11.543
optimal	7.442	10.986
exhaustive	29.200	29.200

Negative finding: Under these cleaner-environment priors CERA **ties cost-greedy** and does not match the exact small-instance optimum. We also tested a 1/2/3-probe budget; mean decision coverage was respectively 50%, 75%, and 87.5% for every policy on these four test templates. AURA's UNCERTAIN verdict remains explicit under the budgets. There is no claim of superiority on these stress controls.

5. Local integrated verification

The six Stage 2 controlled tasks are: correct completion, missing report, unrelated persistent modification, silent tool no-op, denied authoritative readback, and missing mandatory validation. Each is captured from a new local SQLite/filesystem run. The five methods then inspect the same frozen task evidence and use Stage 4 as the authoritative verifier.

All 30 of 30 results matched expected verdicts: clean PASS, four invalid-state/process examples FAIL, and denied-readback UNCERTAIN. No independent LLM-agent benchmark has yet been used.

Policy	Total probes across six tasks	Mean estimated marginal probe cost (ms)	Mean selection overhead (ms)
fixed	18	17.700	0.029
greedy	18	14.783	0.054
cera	18	14.783	0.861
optimal	18	14.783	2.109
exhaustive	24	26.000	0.017

Cost caveat: The estimates in the table are predefined adapter cost parameters, not measured latency. Measured probe and planner latencies and the common snapshot cost are recorded in local_live_policy_results.csv and individual certificates. On this tiny environment, CERA's additional selection overhead can outweigh probe-time savings. No credible system-level speedup is established.

6. Quality assurance and reproducibility

• **153 automated tests passing**, including 115 inherited regression tests and 38 Stage 5 tests (finite-model planning, coverage, failure-dominance, no hidden-world leakage, exact comparison, budgets, invalid models, real Stage 2 replay, and all 30 integration decisions).

• Stage 4's validated Observation admission and Stage 1 Certificate shape preserved. No direct arbitrary fact injection from the planner.

• Every result includes the selected probes, stepwise state of uncertainty, policy, and costs; live results contain the verification certificate and bounded reasoning ledger.

• reproduce_stage5.py refuses to overwrite nonempty output directories; it regenerates simulations, prior-shift tests, budget controls, and six local execution captures.

Reproduction:

```
python -m pip install -e '.[test]'
python -m pytest -q
python reproduce_stage5.py --out ./fresh_stage5_results
```

The synthetic summaries should reproduce numerically on subsequent runs. Fine-grained wall-clock milliseconds may vary due to machine load.

7. Limitations, threats and required next stage

1. The benchmark has 240 enumerated **world assignments**, not 240 independent workflows. All four domains share closely related Boolean structures. The fault model and successful contract are co-authored; external validity is untested. 2. Four small Boolean obligations do not capture realistic semantics, long-horizon causal provenance or unobserved external effects. 3. The exact optimizer assumes deterministic probe results and a correctly specified prior. It is computationally limited to tiny cases. 4. Original CERA's separation heuristic may be tied or beaten by greedy under other conditions; it is not SOTA or proven optimal. 5. Whole-frame completeness and trustworthy snapshots remain assumptions. No claim covers transient reverted effects or external services outside the declared boundary. 6. Stage 5 does not show statistically significant or practically meaningful advantage over strong heuristics on external tasks. Our earlier aspirational 25% cost improvement target was **not met** against fixed or greedy in this pilot.

Stage 6 priority: adapt to externally sourced tasks and trajectories (AppWorld / Agent-Diff as compatible), add information-matched state-only/trace-only and targeted-evidence baselines, construct held-out fault mixtures with blind ground truth, and evaluate overall capture + planning + inspection costs. Only after that can we judge whether CERA is a research contribution or an illustrative engineering implementation.

Repository: No GitHub commits or pushes were made; the link supplied by the user is not required for this standalone Stage 5 implementation.